

Problem A

ASCII Art Contest

Time limit: 1 second

Problem Description

Three leading AI-powered creative assistants—Gemini, ChatGPT, and Claude—enter the first-ever ASCII Art Contest, where they must impress a panel of human judges with their text-based masterpieces.

Each participant receives a score between 80 and 100 (inclusive). The organizers want to announce the final standings only if the judges' opinions are “close enough”; otherwise, they will ask the judges to reconsider.

Given the three integer scores of Gemini, ChatGPT, and Claude, determine the contest result:

- If the maximum and minimum scores differ by at least 10 points, print **check again** (the judging seems inconsistent, so the panel must re-evaluate).
- Otherwise, print **final X**, where **X** is the median of the three scores (the score that would be in the middle if all three were sorted in nondecreasing order).

Input Format

The only line of input contains three integers g , c , and ℓ , representing the scores of Gemini, ChatGPT, and Claude, respectively.

Output Format

Print the required answer in a line.

Technical Specification

- $80 \leq g, c, \ell \leq 100$

Sample Input 1

```
88 94 95
```

Sample Output 1

```
final 94
```

Sample Input 2

```
100 80 81
```

Sample Output 2

```
check again
```

Sample Input 3

```
98 99 98
```

Sample Output 3

```
final 98
```

Sample Input 4

95 86 85

Sample Output 4

check again

Problem B

Buses

Time limit: 1 second

Problem Description

There is a long straight road of length ℓ meters, where *position* p denotes the point on the road that is p meters away from the starting point. Along this road, there are n buses moving in the positive direction, each traveling at the **same constant speed** of x meters per minute. The i -th bus is currently at position s_i and continues moving until it reaches its designated destination at position t_i . Once a bus reaches its destination, it ceases operation and all passengers must disembark.

There are also m people who wish to reach the end of the road (position ℓ). The current position of the i -th person is p_i , and each person can walk at a speed of **at most** y meters per minute. If a person is at the same position as a bus, they may hop on the bus instantly. While riding a bus, they may hop off at any moment. The time required to board or leave a bus is considered negligible. Buses always move at a constant speed x and never wait for passengers.

Your task is to determine the minimum possible time for each person to reach the end of the road.

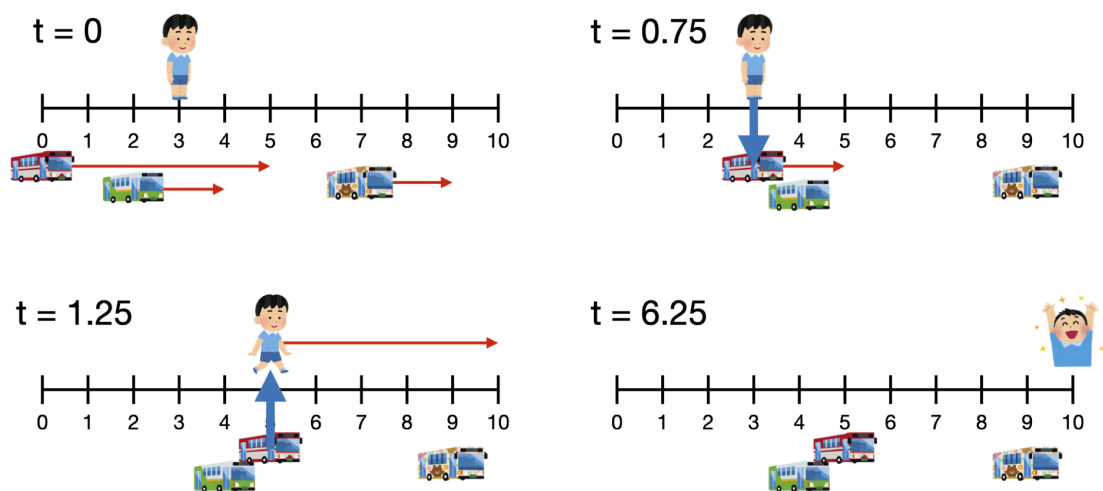


Figure B.1: An illustration for sample input 1.

Input Format

The first line contains five integers n , m , ℓ , x , and y , representing the number of buses, the number of people, the length of the road, the speed of the buses, and the walking speed of the people, respectively.

The i -th of the following n lines contains two integers s_i and t_i , representing the starting position and the destination position of the i -th bus.

The i -th of the following m lines contains one integer p_i , representing the current position of the i -th person.

Output Format

Print m lines. The i -th line contains a number which is the minimum time (in minutes) for the i -th person to reach the end of the road.

Your answer will be accepted if the absolute or relative error does not exceed 10^{-6} . Formally, let your answer be a , and the jury's answer be b . Your answer is considered correct if $\frac{|a-b|}{\max(1,|b|)} \leq 10^{-6}$.

Technical Specification

- $1 \leq n \leq 2 \times 10^5$
- $1 \leq m \leq 2 \times 10^5$
- $1 \leq \ell \leq 10^9$
- $1 \leq y < x \leq 10^6$
- $0 \leq s_i < t_i \leq \ell$
- $0 \leq p_i \leq \ell$

Sample Input 1

```
3 3 10 4 1
0 5
2 4
7 9
3
8
5
```

Sample Output 1

```
6.25
1.5
5
```

Explanation of Sample 1: A person initially at position $p = 3$ can reach the end of the road in 6.25 minutes as follows:

- Wait for Bus 1 to arrive.
- Hop on the bus and ride it until it reaches its destination at position $t_1 = 5$.
- Get off the bus and walk the remaining distance to position $\ell = 10$.

As shown in Figure B.1, the total time spent is 6.25 minutes, which is the minimum possible.

Sample Input 2

```
1 3 100 100 1
1 2
0
1
2
```

Sample Output 2

```
100
98.01
98
```

This page is intentionally left blank.

Problem C

Circles Are Far from Each Other

Time limit: 3 seconds

Problem Description

You are given n circles and two integer parameters k and ℓ . The i -th circle has radius r_i and $r_i \geq r_j$ holds for every $1 \leq i < j \leq n$. The task is to draw these n circles on a 2D plane so that the following conditions are simultaneously satisfied. Before proceeding to the conditions, let us recall that:

- Each circle is uniquely determined by its center O and radius r .
- A circle of radius r is defined as the set of all points whose distance from its center O is **exactly** r . All distances considered in this task are Euclidean.
- The interior region of a circle is defined as the set of all points whose distance from its center O is **less than** r .
- We say that one circle C *encloses* another circle C' if all points of C' lie inside the interior region of C .

You need to draw these n circles to satisfy all of the following conditions:

1. The centers of all circles are collinear.
2. The distance between any two centers is at most k .
3. No two circles intersect.
4. If a circle encloses two circles C and C' , then either C encloses C' or C' encloses C .
5. The first ℓ circles **may or may not** be enclosed by other circles, whereas each of the remaining $n - \ell$ circles **must** be enclosed by at least one circle.

An arrangement that satisfies all the above conditions is called *feasible*. For a feasible arrangement $\mathcal{A}$, define its *quality* $d(\mathcal{A})$ as the minimum distance between any two points belonging to different circles.

If there exists at least one feasible arrangement for the given case, output the maximum possible quality among all feasible arrangements. If no feasible arrangements exist, output 0.

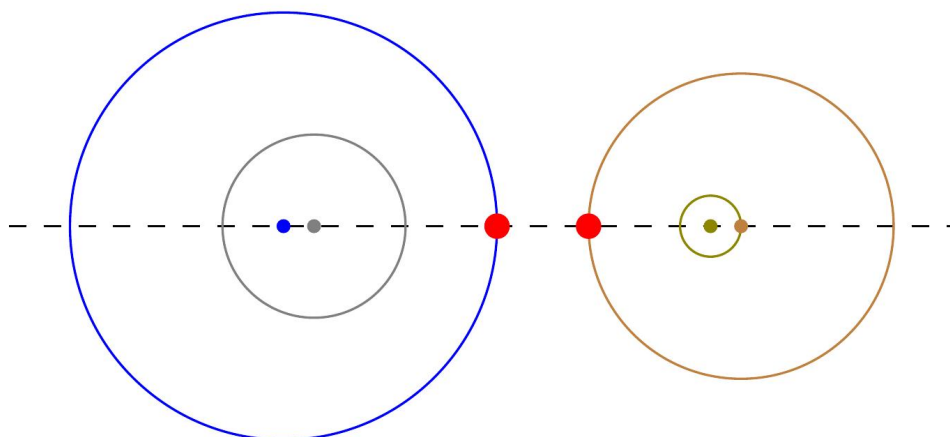


Figure C.1: A feasible arrangement for sample input 1.

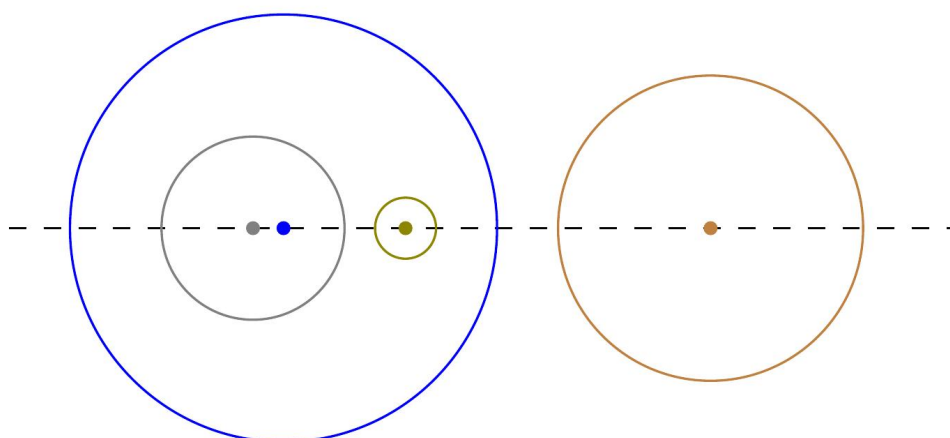


Figure C.2: An arrangement that is **not feasible** for sample input 1.

Input Format

The first line contains three integers k , n , and ℓ , representing the maximum distance between centers, the number of circles to be drawn, and the number of circles that may or may not be enclosed by other circles, respectively.

The second line contains n integers $r_1, r_2, \dots, r_n$, where r_i is the radius of the i -th circle.

Output Format

If no feasible arrangements exist, output a single integer 0.

Otherwise, output the maximum possible quality $d(\mathcal{A})$ among all feasible arrangements $\mathcal{A}$ in the following format:

- If $d(\mathcal{A})$ is an integer, output it as a single integer.
- If $d(\mathcal{A})$ is not an integer, output it in the rational form a/b such that $1 \leq a, b \leq 10^9$, $\gcd(a, b) = 1$, and $|d(\mathcal{A}) - a/b|$ is minimized. If multiple such forms satisfy the constraints, print any.

Technical Specification

- $1 \leq k \leq 10^9$
- $2 \leq n \leq 10^5$
- $1 \leq \ell \leq \min\{n, 200\}$
- $1 \leq r_i \leq 10^9$
- $r_i \geq r_j$ for every $1 \leq i < j \leq n$.

Sample Input 1

```
15 4 3
7 5 3 1
```

Sample Output 1

```
3
```

Explanation of Sample 1: A feasible arrangement is illustrated in figure C.1, where each circle and its center are shown in a unique color.

In this arrangement, the minimum distance between any two points belonging to different circles is 3, and the two points witnessing this distance are marked as red dots. Hence, the quality of this arrangement is 3, which is the maximum possible among all feasible arrangements for this case. Note that only circle 4 is required to be enclosed within another circle, while the remaining circles may or may not be enclosed.

Figure C.2 shows an arrangement that is **not feasible**. In this case, circle 1 encloses both circle 3 and circle 4, but neither of these two circles encloses the other, thereby violating condition 4.

Sample Input 2

```
14 6 1
7 5 4 3 2 1
```

Sample Output 2

```
1
```

Sample Input 3

```
14 2 2
5 4
```

Sample Output 3

```
5
```

Sample Input 4

```
22 4 4
4 4 1 1
```

Sample Output 4

```
10/3
```

Sample Input 5

```
13 3 3
6 3 1
```

Sample Output 5

```
4
```

Problem D

Divisor Card Game

Time limit: 1 second

Problem Description

In Taiwan, many mathematics teachers design board and card games to help students grasp difficult mathematical concepts. Recently, a particular card game has gone viral among elementary and middle school teachers because it effectively helps students understand the concepts of divisors and multiples, while also being highly engaging for both teachers and students.

The rules of the game are as follows. The teacher prepares n distinct cards, labeled from 1 to n . The i -th card has an integer value a_i written on it, and the integers $a_1, a_2, \dots, a_n$ are in a strictly increasing order.

There are m students labeled from 1 to m participating in the game. Before the game begins, each student receives a nonempty subset of the n cards. No two students share any card, and at least one card remains undealt.

Let k denote the number of undealt cards initially. The game consists of k rounds. In each round, the following steps occur in order:

1. The teacher selects one of the remaining undealt cards uniformly at random and reveals it to all students. Let c be the integer written on this card.
2. Each student simultaneously chooses exactly one card from their own collection.
3. The ownership of the revealed card is determined as follows:
 - Among the values of all cards chosen by the students, consider those that are divisible by c .
 - If there are one or more such values, the student who selected a card with the **smallest** divisible value wins the revealed card and adds it to their collection.
 - If no chosen cards have value divisible by c , the revealed card is discarded (remains unowned). Discarded cards are not used in subsequent rounds.

Each student follows the same strategy throughout the game:

- If the student owns at least one card whose value is divisible by c , they choose the card with the **smallest** value.
- Otherwise, they choose the card with the **smallest** value among those they own.

Assuming that all students follow this strategy in every round, determine the expected number of cards that each student will own at the end of the game.

Input Format

The first line contains two integers n and m , representing the number of cards and the number of students, respectively.

The second line contains n integers $a_1, a_2, \dots, a_n$, where a_i is the integer on the i -th card.

The third line contains n integers $b_1, b_2, \dots, b_n$, where b_i describes the ownership of the i -th card before the game begins: it is owned by the b_i -th student if $b_i \neq 0$, and is undealt otherwise.

Output Format

Print m numbers in a new line, where the i -th number represents the expected number of cards the i -th student will own at the end of the game.

It can be proven that each answer can be represented by a rational number $\frac{p}{q}$ where q is not a multiple of 998244353. Therefore, you are asked to print $p \times q^{-1}$ modulo 998244353 for each number, where q^{-1} means the multiplicative inverse of q modulo 998244353.

Technical Specification

- $1 \leq m < n \leq 600$
- $1 \leq a_i \leq 10^{18}$
- $0 \leq b_i \leq m$
- $a_i < a_{i+1}$ for all $1 \leq i < n$.
- There is at least one i such that $b_i = j$ for $j = 0, 1, \dots, m$.

Sample Input 1

```
5 2
1 2 3 4 5
0 0 1 2 1
```

Sample Output 1

```
499122179 499122179
```

Sample Input 2

```
6 2
1 2 3 4 5 6
0 0 0 1 0 2
```

Sample Output 2

```
831870297 166374061
```

Sample Input 3

```
8 3
4 5 8 9 12 18 20 24
0 0 0 0 0 2 1 3
```

Sample Output 3

```
332748120 2 665496239
```

Problem E

Number Maze

Time limit: 1 second

Problem Description

At the “Number Maze,” a mysterious code master stands guard. He blocks the path and smiles, saying: “Brave adventurer, passing through this gate will not be easy! An ancient numeric code, capable of being rearranged into numerous combinations, is in the code master’s possession. The traveler must choose two numeric codes from these combinations and show their $xAyB$ result, or the gate will trap the traveler here forever!”

The rules of $xAyB$ are as follows:

- Each A indicates that a digit in both codes matches in both value and position.
- Each B indicates that a digit in both codes matches in value but not position.

For example:

Codes Compared	Result	Explanation
5234 vs. 5789	$1A0B$	Only the digit 5 matches in both value and position.
5634 vs. 6589	$0A2B$	Digits 5 and 6 match only in value but are in different positions.
1847 vs. 6149	$1A1B$	The digit 4 matches in both value and position, while the digit 1 only matches in value.

You are given a base numeric code n , which can be one of $\{12, 123, 1234\}$. Consider all possible permutations of the digits of n , sorted in ascending order. Let the j -th and k -th permutations (1-indexed) be the two numeric codes chosen by the traveler.

Your task is to compare these two permutations and determine their $xAyB$ result, according to the rules above.

Input Format

Each test contains multiple test cases. The first line contains a single integer t , representing the number of tests the code master will conduct. The description of the test cases follows.

The only line of each test case contains three integers n , j , and k , representing the base numeric code and the indices of the two permutations to be compared, respectively.

Output Format

For each test case, output the result in the format $xAyB$, where x and y are integers.

Technical Specification

- $1 \leq t \leq 1000$
- $n \in \{12, 123, 1234\}$
- Both j and k are valid indices of permutations of the digits of n .

Sample Input 1

```
3
12 1 2
123 1 2
123 2 5
```

Sample Output 1

```
0 A2B
1 A2B
1 A2B
```

Sample Input 2

```
3
1234 15 9
1234 1 24
1234 1 1
```

Sample Output 2

```
2 A2B
0 A4B
4 A0B
```

Problem F

Cluster Computing System

Time limit: 1 second

Problem Description

The ICPC company plans to build a cluster computing system consisting of n servers. Each server has a *database protocol type* represented by a positive integer. Specifically, the i -th server has a protocol type p_i .

Initially, all servers are independent. The company wants to establish connections between servers so that, in the resulting network, every server can reach every other server (either directly or indirectly).

To achieve full connectivity, you may establish several connections. Each time you establish a connection, you must choose two servers u and v ($u < v$). The cost of establishing the connection between u and v is defined as the *common protocol* of the databases in the range from u to v , calculated as $\gcd(p_u, p_{u+1}, \dots, p_v)$ ¹.

Determine the minimum total cost required to fully connect all n servers so that every server is reachable from every other server.

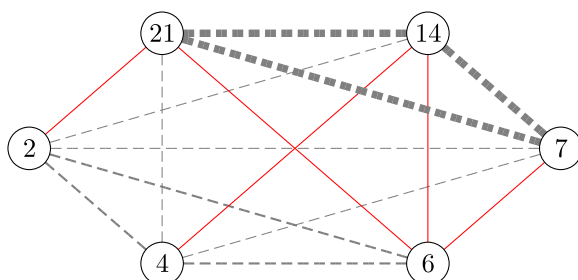


Figure F.1: An illustration for sample input 2.

Input Format

The first line contains a single integer n , representing the number of servers.

The second line contains n positive integers $p_1, p_2, \dots, p_n$, where p_i is the database protocol type of the i -th server.

Output Format

Output a single integer in a line, representing the minimum total cost required to fully connect the cluster computing system.

¹ $\gcd$ is the largest positive integer that divides every integer in that set without leaving a remainder.

Technical Specification

- $2 \leq n \leq 2 \times 10^5$
- $1 \leq p_i \leq 10^9$

Sample Input 1

```
3
4 2 6
```

Sample Output 1

```
4
```

Sample Input 2

```
6
2 4 6 7 14 21
```

Sample Output 2

```
5
```

Explanation of Sample 2: Figure F.1 shows an example with protocol types 2, 4, 6, 7, 14, and 21. The costs of all possible connections are 1, 2, or 7, indicated by different widths. There is a way to connect all together with five connections of cost 1 each (displayed in red).

Problem G

Gene Editor

Time limit: 1 second

Problem Description

Biologists have recently discovered an interesting phenomenon in a particular type of organism. Each organism possesses a gene sequence consisting exclusively of two types of genes, represented by the characters A and B.

These organisms reproduce asexually, meaning that the offspring usually inherit an identical gene sequence. However, due to occasional cloning errors during reproduction, mutations may occur. Biologists have observed that these errors can take the following forms:

1. Inserting the substring AA at any position in the gene sequence.
2. Removing the substring AA from any position in the gene sequence; the remaining parts are concatenated without altering their order.
3. Inserting the substring BBB at any position in the gene sequence.
4. Removing the substring BBB from any position in the gene sequence; the remaining parts are concatenated without altering their order.
5. Inserting a special substring s at any position in the gene sequence.
6. Removing the substring s from any position in the gene sequence; the remaining parts are concatenated without altering their order.

These mutations may occur multiple times during a single cloning event and always happen sequentially, one at a time.

For example, suppose $s = \text{ABAB}$. An organism with gene sequence **ABBABBA** could produce an offspring with gene sequence **A** through the following series of mutations:

```

ABBABBA  ———— insert ABAB ———→ ABBABBAABAB
                ———— remove AA ———→ ABBABBBAB
                ———— remove BBB ———→ ABBAAB
                ———— remove AA ———→ AB BB
                ———— remove BBB ———→ A

```

The biologists possess an organism with a specific gene sequence t . They are also interested in all possible organisms whose gene sequences have length n . Since each position in a gene sequence can be either A or B, there are 2^n such organisms in total.

The question is: Given strings s , t , and n , how many of these 2^n organisms (that is, all gene sequences of length n) can be produced from the organism with gene sequence t through a sequence of valid mutation operations as described above?

Since the answer may be very large, output the result modulo 998244353.

Input Format

Each test contains multiple test cases. The first line contains the number of test cases q . The description of the test cases follows.

The first line contains the string s , representing the special substring.

The second line contains the string t , representing the given gene sequence.

The third line contains an integer n , representing the interested length of organisms.

Output Format

For each test case, print an integer in one line, representing the answer modulo 998244353.

Technical Specification

- $1 \leq q \leq 10$
- $2 \leq |s| \leq 11$
- The first character in s is A.
- The last character in s is B.
- There are no two consecutive A's in s .
- There are no three consecutive B's in s .
- $1 \leq |t| \leq 10^5$
- $1 \leq n \leq 10^9$

Sample Input 1

```
2
ABAB
AABAB
1
ABAB
A
7
```

Sample Output 1

```
1
22
```

Problem H

Shuffling Cards with Problem Solver 68!

Time limit: 2 seconds

Problem Description

Aru loves playing card games (Poker, Texas hold 'em, Balatro, etc.) and she has perfected the art of shuffling cards, especially the *riffle* shuffle. She is playing with Mutsuki now, and it's her turn to shuffle the cards!

However, Mutsuki knows that Aru is too perfect with her shuffling game. In fact, given a deck with an **even** number of cards, Aru always performs a perfect riffle: she cuts the deck evenly and interleaves the two halves. Formally, if the deck is represented by a string s of length n , where s_i is the i -th card from the top, one riffle produces the deck

$$s' = s_1 + s_{n/2+1} + s_2 + s_{n/2+2} + \dots + s_{n/2} + s_n.$$

Mutsuki also knows that when handed a deck of cards, Aru will riffle it exactly t times.

Mutsuki currently holds a deck of 2^k cards, represented by a string d . **Before** giving the deck to Aru, Mutsuki can choose to *cut* the deck, by moving some number of cards from the top to the bottom of the deck. Formally, she can choose any m from 0 to $2^k - 1$, and produce the deck

$$d' = d_{m+1} + d_{m+2} + \dots + d_{2^k} + d_1 + d_2 + \dots + d_m.$$

Among all 2^k possible cuts, Mutsuki wants to choose the one that results in the **lexicographically smallest** deck after Aru riffles it t times. Can you figure this out for her?

Input Format

The first line contains two integers k and t , representing the size parameter of the deck, and the number of times Aru will riffle the deck, respectively.

The second line contains a string d of 2^k lowercase characters, representing the original deck of cards that Mutsuki has.

Output Format

Print a string in one line, representing the lexicographically smallest deck of cards that Mutsuki can produce, by first cutting the deck and letting Aru riffle it t times.

Technical Specification

- $1 \leq k \leq 18$
- $0 \leq t \leq 10^9$

Sample Input 1

```
4 2
baaabaabaabaabaaa
```

Sample Output 1

```
aaaaaaaaaaaaabbbb
```

Sample Input 2

```
4 999999999
abcdefghijklmnop
```

Sample Output 2

```
acegikmobdfhjlnp
```

Sample Input 3

```
4 17
bbcttckrdezzzbcd
```

Sample Output 3

```
bcckdrbdbeczttztz
```

Problem I Birthday

Time limit: 1 second

Problem Description

Anon and Soyo are good friends. To celebrate Anon's birthday, Soyo decides to buy a cake. After careful selection, she chooses a round strawberry cake to share with Anon at home.

The cake is modeled as a circle centered at the origin $(0, 0)$ with radius r . There are n strawberries on the cake, where the i -th strawberry is located at (x_i, y_i) . The distance of any strawberry from the origin is at most $0.9r$.

Soyo wants to cut the cake into two pieces with a single straight line. Since Anon loves strawberries, Soyo wants Anon's piece contains all of them. If a strawberry lies on the cutting line, Soyo can assign it to either piece.

Soyo wants to make the smaller piece as large as possible. Please help Soyo determine the maximum possible area of the smaller piece, given that all strawberries lie on the same piece.

Input Format

The first line contains two integers n and r , representing the number of strawberries on the cake and the radius of the cake, respectively.

The i -th of the following n lines contains two integers x_i and y_i , representing the coordinates of the i -th strawberry.

Output Format

Print a single real number in one line, representing the maximum possible area of the smaller piece, given that all strawberries lie on the same piece.

Your answer will be accepted if the absolute or relative error does not exceed 10^{-6} . Formally, let your answer be a , and the jury's answer be b . Your answer is considered correct if $\frac{|a-b|}{\max(1, |b|)} \leq 10^{-6}$.

Technical Specification

- $1 \leq n \leq 2 \times 10^5$
- $1 \leq r \leq 10^6$
- $\sqrt{x_i^2 + y_i^2} \leq 0.9r$
- All strawberries are at distinct points.

Sample Input 1

```
4 5
-3 -3
3 -3
-3 3
3 3
```

Sample Output 1

```
11.182380450040
```

Sample Input 2

```
7 15
9 -4
2 -2
8 3
0 4
-6 10
6 6
3 5
```

Sample Output 2

```
353.429173528852
```

Sample Input 3

```
15 15
-4 -1
0 -1
2 -9
0 2
8 1
3 -3
-9 3
8 6
9 7
-9 -1
2 6
-2 7
-10 -8
4 0
-5 -8
```

Sample Output 3

```
168.906562205067
```

Problem J Sliding Tiles

Time limit: 4 seconds

Problem Description

You have a special sliding puzzle played on an $n \times n$ grid. This puzzle is slightly different from standard sliding puzzles: between each pair of adjacent columns, there is a vertical bar of height h_i (for $1 \leq i < n$) positioned at the bottom of the grid. Each h_i indicates how many rows from the bottom this bar extends upwards, and it blocks tile movement between the two columns in those rows.

The grid contains several tiles, each occupying exactly one cell. These tiles can slide freely in the grid unless they are blocked by the grid boundaries, a vertical bar (depending on its height) or another tile.

The puzzle allows two types of tilt operations:

- Tilt right: All tiles slide to the right as far as possible.
- Tilt down: All tiles slide downward as far as possible.

In both operations, all tiles move simultaneously and stop only when blocked by the grid's edge, a bar, or another tile.

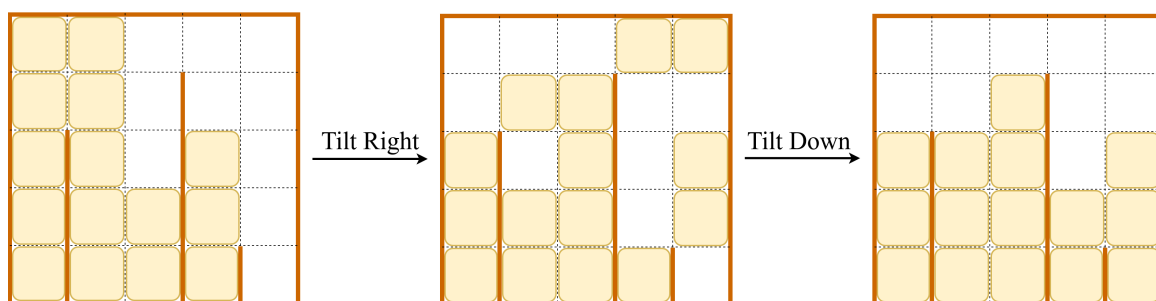


Figure J.1: An illustration for sample input 1.

Define a group operation as a sequence of: first tilt the grid to the right, then tilt it downward.

Initially, the i -th column has a_i tiles stacked from the bottom of the column. You perform the group operation exactly once on the board. After the operation, determine the number of tiles in each column.

Input Format

The first line contains an integer n , representing the size of the board.

The second line contains n integers $a_1, a_2, \dots, a_n$, where a_i is the number of tiles in the i -th column initially.

The third line contains $n - 1$ integers $h_1, h_2, \dots, h_{n-1}$, where h_i is the height of the bar between column i and column $i + 1$.

Output Format

Print n numbers in a new line, representing the number of tiles in each column after performing the group operation exactly once.

Technical Specification

- $2 \leq n \leq 5 \times 10^5$
- $0 \leq a_i \leq n$
- $0 \leq h_i \leq n - 1$

Sample Input 1

```
5
5 5 2 3 0
3 0 4 1
```

Sample Output 1

```
3 3 4 2 3
```

Sample Input 2

```
6
4 3 0 3 0 2
1 0 0 1 3
```

Sample Output 2

```
1 0 3 3 2 3
```

Problem K

Kindergarten Homework

Time limit: 2 seconds

Problem Description

Kenny is a kindergarten teacher who is teaching his students about math. To help them learn, he wants to give them homework in an enjoyable and non-annoying format – Number Search.

A Number Search puzzle is similar to a Word Search puzzle, but instead of words, it's about finding math expressions. An instance of the puzzle consists of two parts: a grid with n rows and m columns, and a list $a_1, a_2, \dots, a_q$ of q target numbers to search for. Each cell in the grid contains a digit from 1 to 9, a plus sign $+$, or a multiplication sign $*$.

An expression can be formed by joining one or more consecutive characters along the same row, column, or diagonal, in a straight line (in either direction). Each expression can be defined by its starting cell and ending cell. Two expressions are considered different if they start or end at different cells, even if they contain the exact same characters.

A valid expression must be of the form

$$num \ op \ num \ op \ \dots \ op \ num$$

where each num is a sequence of one or more digits, and each op is either $+$ or $*$. For example, "123" and "1+2*3+45" are valid expressions, while "+123" and "2**5" are not. The value of an expression is the result of evaluating it following typical arithmetic rules.

The goal of the puzzle is to count, for each number a_i , how many different valid expressions in the grid evaluate to a_i .

To demonstrate the rules, consider the following completed Number Search puzzle:

Number Search!!!

4	2	1	6	7	9	3	+
7	1	7	*	4	*	5	4
7	+	5	+	7	2	7	*
4	5	1	4	9	+	7	1
8	+	2	6	6	9	7	*
+	1	8	9	*	2	+	9
5	+	*	2	4	4	+	7
4	2	5	9	5	9	5	2
9	7	+	*	3	1	5	+

**Numbers
to find**

67

420

3

727

Let's denote the cell at row r and column c as (r, c) .

1. The answer for 67 is 2, as there are two expressions "67" that start at $(1, 4)$. Although both expressions contain the same characters, they are considered different because they end at different positions.
2. The answer for 420 is 0, as no expressions in the grid evaluate to 420.
3. The answer for 3 is 4:
 - The expression "3" appears twice, at $(1, 7)$ and $(9, 5)$.
 - The expression "1+2" appears from $(6, 2)$ to $(8, 2)$.
 - The expression "2+1" appears from $(8, 2)$ to $(6, 2)$. Even though they share the same set of cells, the two expressions are considered different because they start at different positions.
4. The answer for 727 is 3:
 - The expression "7+16*45" appears from $(2, 1)$ to $(8, 7)$. Note that multiplication is performed before addition.
 - The expression "727" appears from $(3, 5)$ to $(3, 7)$, and from $(3, 7)$ to $(3, 5)$. Even though they share the same set of cells and contain the same characters, they are considered different because they start at different positions.

Kenny can't wait to see children enjoy solving interesting Number Search puzzles. But before giving them a puzzle, he needs to have the answers ready. Please help him calculate the answers for each puzzle he has prepared.

Input Format

The first line contains three integers n , m , and q , representing the number of rows, the number of columns, and the number of target numbers, respectively.

Each of the following n lines contains a string of m characters, representing the grid.

The i -th of the following q lines contains an integer a_i , representing the i -th target number.

Output Format

Print q lines. The i -th line contains an integer, representing the answer for a_i .

Technical Specification

- $1 \leq n, m \leq 1000$
- $1 \leq n \times m \leq 3 \times 10^4$
- $1 \leq q \leq 10^5$
- The grid contains only the characters in “123456789+*”.
- $1 \leq a_i \leq 10^6$
- All a_i are distinct.

Sample Input 1

```
9 8 4
4216793+
717*4*54
7+5+727*
45149+71
8+26697*
+189*2+9
5+*244+7
42595952
97+*315+
67
420
3
727
```

Sample Output 1

```
2
0
4
3
```

This page is intentionally left blank.

Problem L

Maximum Color Segment

Time limit: 2 seconds

Problem Description

You are given a rope n units long, where each unit is painted either red or black. The rope can be represented as a string of length n consisting of characters R (red) and B (black). You are also given two positive integers m and k .

You may perform the following operation **at most** m times (possibly, zero times):

- Choose any contiguous substring² of the rope of length exactly k .
- Flip the color of every unit in the substring: each R becomes B, and each B becomes R.

For example, consider the rope RRRRBRRR with $k = 4$. If you choose the 3-rd to 6-th characters (RRRBRRR), then after flipping, the rope becomes RRBBRBR.

Define the *number of color segments* as the smallest number of contiguous segments into which the rope can be divided so that each segment consists of units of a single color. For instance, the rope RRRRRBBB has 4 color segments: RR, B, RRR, and BB.

Your task is to determine the maximum possible number of color segments after performing **at most** m operations.

Input Format

The first line contains three integers n , m , and k , representing the length of the rope, the maximum number of operations allowed, and the length of each operation's flip window, respectively.

The second line contains a string of length n consisting only of the characters R and B, representing the initial content of the rope.

Output Format

Output an integer in a single line, representing the maximum possible number of color segments after performing at most m operations.

²A string t is a substring of a string s if t can be obtained from s by the deletion of several (possibly, zero or all) characters from the beginning and several (possibly, zero or all) characters from the end.

Technical Specification

- $1 \leq n \leq 3000$
- $0 \leq m \leq 3000$
- $1 \leq k \leq n$

Sample Input 1

```
5 4 3
RRBRR
```

Sample Output 1

```
5
```

Sample Input 2

```
10 3 3
RRRRBBRRRB
```

Sample Output 2

```
8
```

Sample Input 3

```
7 4 7
RRBRBBR
```

Sample Output 3

```
5
```


Technical Specification

- $1 \leq n \leq 2 \times 10^5$
- $0 \leq m \leq 2 \times 10^5$
- $1 \leq k \leq n$
- $1 \leq a_i \leq k$
- $1 \leq u_i, v_i \leq n$
- The graph is connected and simple (no self-loops or multiple edges).
- Every product type from 1 to k is produced by at least one city.

Sample Input 1

```
3 3 2
2 1 1
1 2
3 1
3 2
```

Sample Output 1

```
1 0
```

Sample Input 2

```
8 10 5
5 1 1 2 4 4 2 3
8 2
4 6
6 2
5 2
2 7
8 6
1 2
7 6
3 1
1 5
```

Sample Output 2

```
1 3 2 2 0
```

Explanation of Sample 2: Figure M.1 illustrates this example. In this example, the port is the blue node. All goods of product type 5 are already in the port, so their transportation cost is 0. The transportation cost for product type 4 is higher. One city with product type 4 has a distance of 1 to the port, while another one has distance 2. Therefore the transportation cost is 2.

Problem N

New Kingdom

Time limit: 1 second

Problem Description

In the upcoming strategy game *Island Cartographer: Penultimate Chapter* (ICPC), players take the role of bridge engineers employed by the *Imperial Cartography & Planning Council* to design a new kingdom spread across an archipelago of n islands connected by a network of causeways. To achieve balance between stability and exploration, the council requires that the player's plan satisfy several strict rules.

The first and fundamental requirement is that the kingdom must be **simple** and **connected**. That is, each causeway connects two different islands, no two causeways connect the same pair of islands, and for each pair of different islands, there exists a route between them using a sequence of causeways.

After learning the ancient wisdom of the legendary tourist Euler, some visitors might be able to traverse every causeway exactly once in a single, seamless tour. If this is possible, then those visitors would not have enough time to enjoy their trips. To make them travel for longer, the council's second requirement is that **exactly** k islands have an odd number of causeways connected to them.

Some causeways are considered **vital** to the kingdom. A causeway is vital if removing it would make travel between some pair of islands impossible. These causeways are important for local development, cultural identity, and tourism, but having too many of them could lead to traffic congestion. Therefore, the council's third and final requirement is that the kingdom must contain **exactly** b vital causeways.

To complete a game of ICPC, the player must submit a plan to build causeways that satisfies all the requirements for the parameters n , k , and b provided by the council. Before the game's release, you are tasked with testing the game by designing a valid kingdom plan as the player. In case the game is impossible to complete, please report so.

Input Format

Each test contains multiple test cases. The first line contains the number of test cases t . The description of the test cases follows.

The only line of each test case contains three integers n , k , and b , representing the number of islands, the exact number of islands connected to an odd number of causeways, and the exact number of vital causeways, respectively.

Output Format

For each test case, if there are no solutions that fit the requirements for a valid plan, output “No” in a single line.

Otherwise, output “Yes” in the first line. Then, output a single integer m in the second line, indicating the number of causeways in your plan. In each of the next m lines, output two integers u_i and v_i separated by a whitespace, meaning that the islands u_i and v_i are connected with a single causeway. If there are multiple valid plans, output any.

See the Technical Specification section for constraints on the output.

Technical Specification

- $1 \leq n \leq 10^5$
- $0 \leq k \leq n$
- $0 \leq b \leq n - 1$
- The sum of n over all test cases does not exceed 10^5 .

Your output must satisfy:

- $0 \leq m \leq 5n$
- For all $1 \leq i \leq m$, $1 \leq u_i, v_i \leq n$.
- For all $1 \leq i \leq m$, $u_i \neq v_i$.
- For all $i \neq j$, $\{u_i, v_i\} \neq \{u_j, v_j\}$.

It can be proven that for every possible input that has a solution, there exists a solution that satisfies these constraints.

Sample Input 1

```
4
6 2 1
3 1 2
4 0 0
5 4 2
```

Sample Output 1

```
Yes
7
1 2
1 3
2 3
3 4
4 5
4 6
5 6
No
Yes
4
1 2
2 3
3 4
4 1
Yes
5
1 4
1 2
4 5
5 1
5 3
```